



Zero Waste: Future of Nepal

Abiral Giri

Saint Xavier's College, Maitighar, Kathmandu, Nepal

Advisor: Mr. Birendra Kumar Jha

Nightingale International Secondary School, Kupondole, Lalitpur, Nepal

1.Introduction

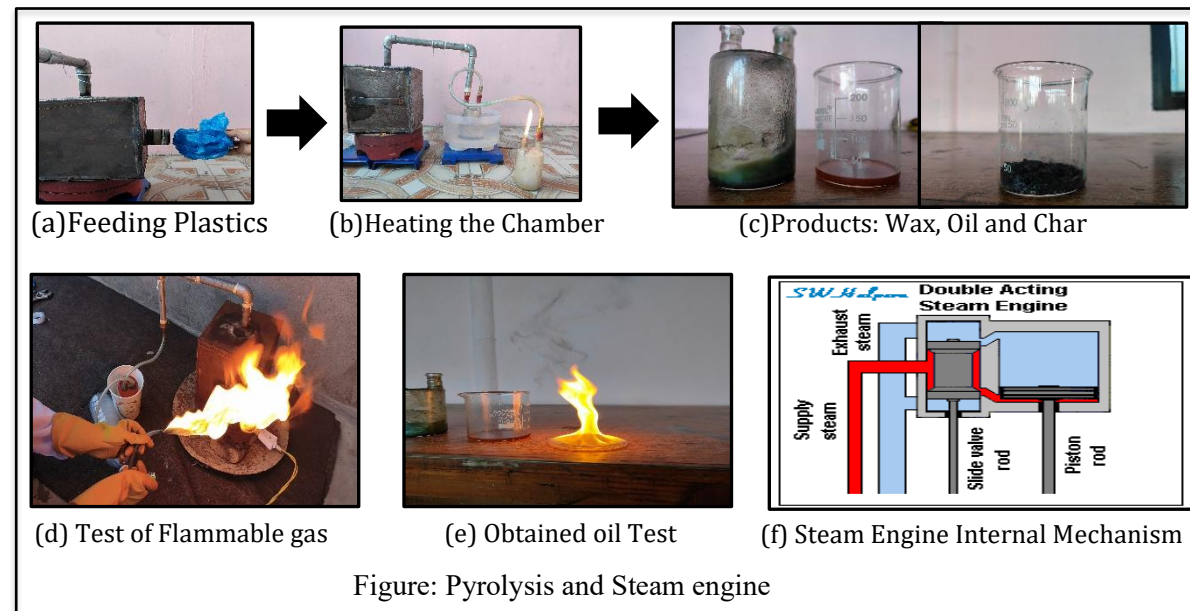
In the current scenario, waste management has become a major challenge that needs to be sorted out to ensure a better and sustainable future for humans. Plastic has limitless potential but once used, it is very difficult to manage and may take up to 500 years to decompose. (Isoto et al.,2019) So, plastics that do not produce toxic/ carcinogenic (cancer causing) fumes High Density Polyethylene (HDPE), Low Density Polyethylene (LDPE), Polypropylene (PP) and Poly styrene (PS) can be separated and sent to the pyrolysis plant. The plastics that do produce carcinogenic and corrosive fumes Poly-Vinyl Chloride (PVC), Poly-Ethylene Terephthalate (PET), etc. (Lopez et al. (2012)) can be dumped into organized landfills and soil, water and air pollution can be minimized. Organic waste is another challenge because of its huge quantity and is being dumped into landfills which causes foul smell and produces leachate by reacting with other chemicals that can cause soil pollution and pollute the underground water resources. Therefore, biogas plants can be constructed in every municipality that can generate useful biogas along with organic fertilizers that can be used for increasing crop yield. This study is about how we can efficiently manage and convert the organic and plastic waste into fuel and other useful resources and use the heat produced to heat water and generate electricity through steam engines.

Keywords: *Municipal solid wastes, Plastic pyrolysis, Fuel, Biogas, Organic fertilizers, Electricity*

2.Methodology

2.1. Pyrolysis of LDPE plastics (Small scale)

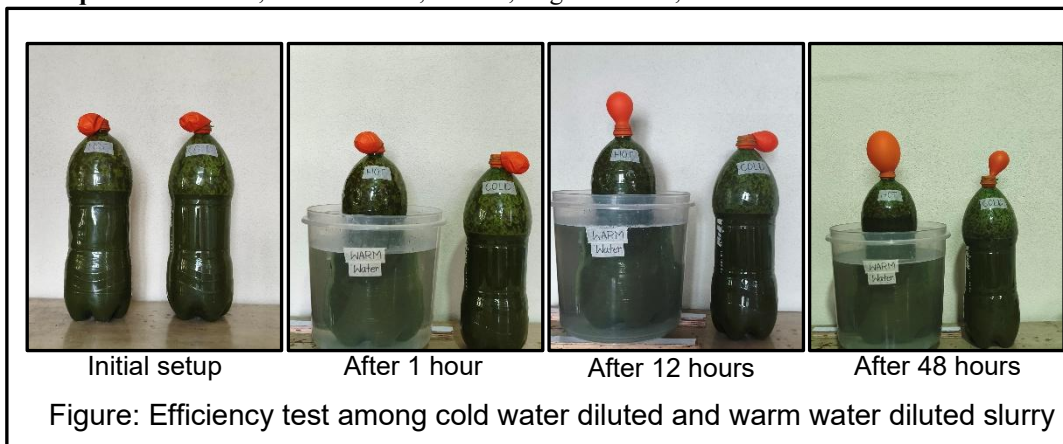
Materials Required: Pyrolysis chamber, Metal pipes, Glass beaker, Water trough, Corks, I- connectors, Heater





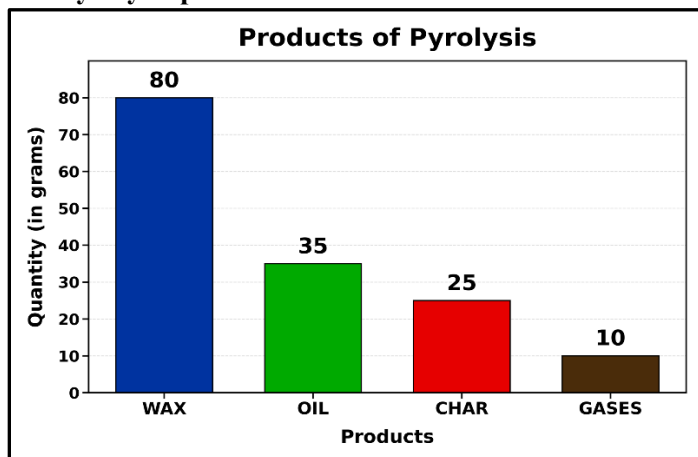
2.2. Two stage biogas system (Small scale)

Materials Required: Balloons, Rubber bands, Bucket, Organic waste, Water heater



3.Results and Discussion

3.1. Pyrolysis products

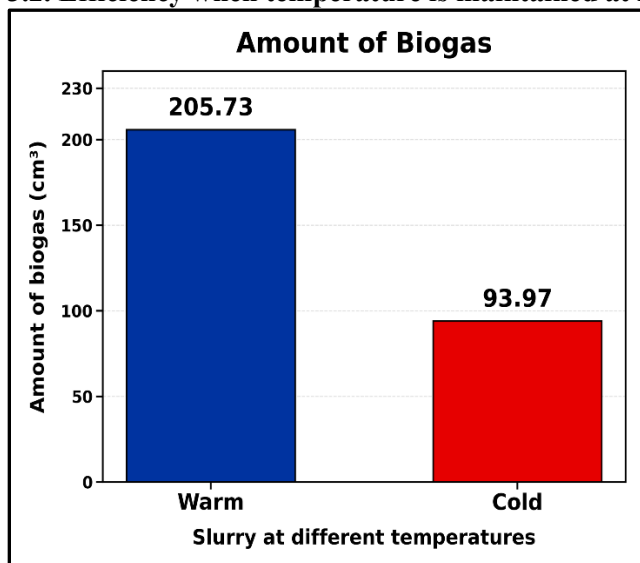


Discussion:

Pyrolysis of about 150 g of LDPE plastics was performed. The formation of was significantly high and that of the fuel was low. This is because the vessel made through welding metal sheets was not completely airtight, purging was not possible to perform, also the heater used to heat the chamber could not heat the chamber above 400°C and catalysts were not used to make the cracking efficient. If these processes are followed and a proper setup is built, the oil yield will be above 70 to 80% of the original mass of the plastics. (Shah et al. (2010))



3.2. Efficiency when temperature is maintained at mesophilic range in biogas plant



Shape of inflated balloons-Prolate Spheroid For the hot water system:

$$\text{Polar radius } (c) = \frac{\sim 9.3 \text{ cm}}{2} = \sim 4.65 \text{ cm}$$

$$\text{Equatorial radius } (a) = \frac{\sim 6.5 \text{ cm}}{2} = \sim 3.25 \text{ cm}$$

$$\text{Volume } (V_1) = \frac{4}{3} \pi a^2 c = 205.73 \text{ cm}^3$$

For the cold-water system:

$$c = \frac{\sim 6.9}{2} = \sim 3.45 \text{ cm}$$

$$a = \frac{\sim 5.1 \text{ cm}}{2} = 2.55 \text{ cm}$$

$$\text{Volume } (V_2) = \frac{4}{3} \pi a^2 c = 93.97 \text{ cm}^3$$

$$\text{Efficiency} = \frac{V_1 - V_2}{V_2} \times 100\% = 118.93\%$$

Discussion:

In this small-scale model, keeping the water warm for 1st bottle was not possible to do 24/7. If the steady temperature is maintained and if thermophilic sweet spot is achieved, the speed efficiency of the system will be well above 300%.

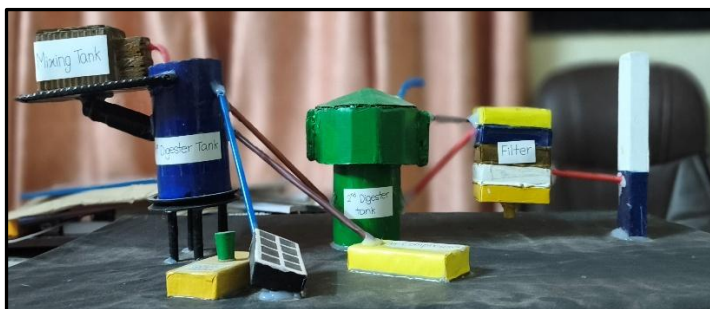


Figure: Two stage biogas plant model

4. Conclusion

These systems can efficiently transform Nepal's wastes into valuable resources. Pyrolysis of plastics and double stage decomposition of organic wastes can reduce about 80% of the Municipal Solid Wastes that would otherwise end up in the unorganized landfills. Thus, preventing environmental pollution while also producing many valuable resources.

5. Acknowledgement

I would like to express my sincere gratitude towards St. Xavier's College and Nightingale International Secondary School for their support. I would also like to thank Mr. Birendra Kumar Jha, Dr. Prashamsa Panta and Dr. Elina Maharjan for their guidance throughout this journey.

6. References

1. Miandad, R., Barakat, M. A., Aburizaiza, A. S., Rehan, M., & Nizami, A. S. (2016). Catalytic pyrolysis of plastic waste: A review. *Process Safety and Environmental Protection*, 102, 822–838. <http://dx.doi.org/10.1016/j.psep.2016.06.022>
2. Istoto, E. H., Widayat, W., & Saptadi, S. (2019). Production of fuels from HDPE and LDPE plastic waste via pyrolysis methods. *E3S Web of Conferences*, 125, 14011. <https://doi.org/10.1051/e3sconf/201912514011>
3. Verma, R., & Kiran, S. C. (2025). Evaluation of biogas production from food waste under thermophilic and mesophilic conditions under Swiss conditions. *Environmental Management and Sustainable Development*, 14(1). <https://doi.org/10.5296/emsd.v14i1.22238>